
VATICINUS • PRE-CONSENSUS • DATED • FALSIFIABLE

Where the constraint moves next: six inelastic inputs the market has not priced

Rent accrues to the binding input. These six calls name the specific molecule, isotope, donor body, or tacit craft that cannot scale on its price signal, and we mark exactly where each claim stops.

The frame: when a system scales, the money does not stay where the headlines are. It moves to the one input that cannot scale with it. Our job is to name that input, and the date it starts to bite, before the market prices it in.

Area: cross-domain: critical materials, energy, biotech, AI-for-science, human capital • Horizon: 2030 to 2040

How we work: cast wide for ideas, then test every one hard against the price and the supply chain. We name the specific input, not the theme.

Two numbers per call: how strong we think the case is, and the odds on the exact, dated version, which is the one we are graded on.

Status: the calls that survived our own attempt to break them. Drafted 2026-06-15.

The board: 6 calls that survived the cut

The cross-cutting read

Every call here runs on one mechanism. A downstream system scales fast while one upstream input is fixed by physics, chemistry, demography, or treaty, so the rent migrates onto the input that cannot answer a price signal. Hafnium and sulfuric acid are byproduct-pinned trickles set by someone else's process (zirconium refining, fossil desulfurization), so you cannot make more without overmaking a product nobody wants, and both now face several inelastic buyers converging at once. Lithium-6 is fixed at 7.5 percent of natural lithium and its only volume-proven separation route is treaty-banned, which leaves a China/Russia duopoly two layers below the fusion plasma everyone watches. Blood and the EHV commissioning craft are demographic locks. The donor-eligible cohort and the 1980s-trained energization journeymen are already born and already shrinking while the consuming cohort grows, and you cannot manufacture a young vein or a four-to-seven-year tacit seasoning. Connectome proofreading flips the usual AI story on its head, because a better auto-segmentation model raises the value of the human gold reference it must be checked against rather than erasing it. The honest boundary across all six is the same. The structural mechanism is near-locked, but each dated trigger carries a real timing-and-measurement tax because a demand-side release valve, a crash program, or a wording threshold can blunt the exact resolution without touching the physics. We think the edge is the input, not the headline, and the discipline is keeping the trigger odds honest.

At a glance

#	THE BOOM	BINDING CONSTRAINT (THE NEEDLE)	CASE	OUR CALL	RESOLVES
P1	Hafnium has no ore. It comes out only as the poison fraction you strip from nuclear-grade zirconium, because Zr cladding has to be neutron-transparent and Hf is a neutron absorber. The same purification step that cleans the cladding concentrates the hafnium, at a fixed ratio of about one tonne of Hf per fifty tonnes of Zr. Over 90 percent of world hafnium arrives this way, from a few qualified separation plants making a few hundred tonnes a year, with end-of-life recycling near 1 percent. Supply cannot answer a hafnium price signal: to make more poison you	Nuclear- and electronic-grade high-purity hafnium: the poison fraction stripped off nuclear-grade zirconium sponge, spec	82%	70%	2032-12-31

#	THE BOOM	BINDING CONSTRAINT (THE NEEDLE)	CASE	OUR CALL	RESOLVES
	would have to overproduce reactor cladding the market does not need. Three demand curves are now converging onto that fixed trickle. Control rods, where roughly half of hafnium demand already sits and the SMR and reactor buildout adds more. Chips, where HfO2 is the high-k gate dielectric at every advanced logic node and the gate-all-around plus AI-compute ramp raises hafnium per wafer, with leading-edge research moving to hafnium-zirconium superlattices that use even more of it. And gas-turbine and aero blades, where hafnium is the metallurgically specific creep-strength additive in single-crystal nickel superalloys, the same turbines being pulled to power AI datacenters. China curbed exports in the second half of 2025. Unwrought hafnium shipments fell about 90 percent, from roughly 5,000 kg in January to under 500 kg by September, and the price tripled.				
P2	Auto-segmentation models trace neurons, but every merge or split is a silent topology error until a human checks it against ground truth, and the model cannot verify itself because the model is what is being checked. As imaging throughput and model quality rise, the gate moves to the one input that does not scale: a human-proofread reference connectome. A	Expert-proofread, error-audited gold-standard connectome reference data (synapse-level verified wiring) plus the human a	82%	46%	2034-12-31

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	mouse brain holds roughly 70-100M neurons and on the order of 100B-1T synapses, each a potential silent error resolved at a roughly fixed human rate. The Wellcome roadmap and independent estimates put whole-mouse-brain proofreading at 0.3-1M person-years and over 90-97% of project cost (about \$10B over 17 years). MICrONS and FlyWire are the existence proofs, and the larval zebrafish reconstruction is still being proofread years after imaging. The trainable expert pool grows linearly at best, and synthetic reference generation is circular: you would train the verifier on the thing it must verify.				
P3	Two decarbonizing systems collide on one molecule that nobody makes on purpose. HPAL nickel-cobalt is roughly 75 percent sulfur-dependent by reagent mass and phosphate fertilizer consumes about 60 percent of global sulfuric acid, yet 80-85 percent of sulfur is an involuntary byproduct of refining sour crude and gas, produced because emissions rules force desulfurization, not because anyone wants it. Burn less fossil fuel and that stream peaks and falls while acid demand climbs toward roughly 400 Mt by 2040 (UCL projects a 100-320 Mt shortfall). Supply is set by oil-and-gas throughput, not by sulfur's own price, so the market cannot	Non-fossil-byproduct or smelter-sourced merchant sulfuric acid available to HPAL and phosphate plants, and the involunta	78%	46%	2035-12-31

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	clear by incentivizing more. The 2026 spike (sulfur up about 70 percent, Huayou cutting HPAL output about 50 percent, Mosaic cutting about 2 Mt of phosphate) is the acute preview. The structural version is the same squeeze with the geopolitics removed and the byproduct base permanently lower.				
P4	The loud, life-or-death labor story is the generic electrician for data-center fit-out, now broadly covered and largely priced into the journeyman wage series. The unpriced input sits one layer up and narrower: the extra-high-voltage transmission craft. Energizing a 345-765 kV line, commissioning an HVDC converter station, or doing live-line work on the bulk grid is a separate, much smaller trade learned almost entirely on the job over many years, with thin apprenticeship pipelines. Live search confirms a roughly 3:1 demand-to-supply gap for HVDC and power-electronics commissioning skill and a "small and shrinking" pool with hands-on EHV-scale experience. You cannot cross-train a data-center journeyman into an EHV energization lead in one apprenticeship cycle. As interconnection queues clear and HVDC links get funded, the schedule binds on this craft, and EHV commissioning day-rates detach from the general electrician market. The live offset to watch: factory	Extra-high-voltage and HVDC transmission commissioning and live-line craft labor: the journeymen, test and protection en	72%	46%	2033-12-31

#	THE BOOM	BINDING CONSTRAINT (THE NEEDLE)	CASE	OUR CALL	RESOLVES
	acceptance testing and digital pre-commissioning are actively pulling work off-site and partly relieving the engineer-side bottleneck.				
P5	When private fusion proves net energy and the plasma bottleneck is funded away, rent jumps upstream into the fuel cycle. Every DT plant breeds its own tritium by hitting Li-6 in the blanket, so each needs a large startup inventory of 6Li-enriched lithium (tens to ~100 t of contained Li-6 per plant, versus natural lithium at only 7.5% Li-6). The detail almost nobody prices: the sole industrially proven route, COLEX, runs on thousands of tonnes of mercury and is banned for new use under the Minamata Convention. The West has essentially zero commercial Li-6 supply, only Russia and China produce it, and China is reportedly just buying from Russia. Whoever first fields a Minamata-compliant, non-mercury Li-6 route at industrial scale captures a chokehold on the entire DT fusion industry.	Qualified blanket-grade enriched lithium-6 (Li-6 enriched well above the 7.5% natural abundance, typically 30-90%) produ	78%	46%	2034-12-31
P6	The math is a demographic identity, not a forecast. People over 65 consume the majority of red-cell units, and that cohort is the already-born boomer wave aging into peak transfusion demand. The donor-eligible band (17-65) is shrinking in absolute terms across Europe, Japan, Korea,	O-negative whole-blood/red-cell donors within the genetically-fixed ~7% share of a shrinking 17-65 donor-eligible cohort	78%	46%	2033-12-31

#	THE BOOM	BINDING CONSTRAINT (THE NEEDLE)	CASE	OUR CALL	RESOLVES
	and parts of the US, with roughly 16M eligible US donors in 2025 falling toward 8.7M by 2060 and demand projected to cross supply around 2027. The non-obvious layer is O-negative. Its ~7% share is fixed by genetics, it is drawn from the same shrinking pool, and it is the default for unmatched emergency use. You cannot manufacture O-neg donors, and the manufactured escape hatches (enzyme-converted and cultured red cells) are still pre-clinical or at milliliter scale through this horizon.				

Case = how strong we think the thesis is. Our call = the probability we put on the exact, dated version landing by the deadline, and the number we are graded on. The gap between them is the cost of pinning a date and a hard threshold, not a lack of conviction.

P1 • Hafnium is a byproduct-pinned trickle, now hostage to three inelastic buyers at once: nuclear control rods, advanced-node HfO₂ gate dielectric, and single-crystal turbine-blade superalloys. China cut exports 90 percent in H2 2025 and the price tripled from \$4,365 to roughly \$13,115/kg by April 2026, but we think the long-horizon lock is the byproduct ratio itself, not the export headline.

The boom: Hafnium has no ore. It comes out only as the poison fraction you strip from nuclear-grade zirconium, because Zr cladding has to be neutron-transparent and Hf is a neutron absorber. The same purification step that cleans the cladding concentrates the hafnium, at a fixed ratio of about one tonne of Hf per fifty tonnes of Zr. Over 90 percent of world hafnium arrives this way, from a few qualified separation plants making a few hundred tonnes a year, with end-of-life recycling near 1 percent. Supply cannot answer a hafnium price signal: to make more poison you would have to overproduce reactor cladding the market does not need. Three demand curves are now converging onto that fixed trickle. Control rods, where roughly half of hafnium demand already sits and the SMR and reactor buildout adds more. Chips, where HfO₂ is the high-k gate dielectric at every advanced logic node and the gate-all-around plus AI-compute ramp raises hafnium per wafer, with leading-edge research moving to hafnium-zirconium superlattices that use even more of it. And gas-turbine and aero blades, where hafnium is the metallurgically specific creep-strength additive in single-crystal nickel superalloys, the same turbines being pulled to power AI datacenters. China curbed exports in the second half of 2025. Unwrought hafnium shipments fell about 90 percent, from roughly 5,000 kg in January to under 500 kg by September, and the price tripled. • Domain: critical-materials / shared upstream byproduct (nuclear + semiconductor + aerospace)

STRUCTURAL CASE

82%

OUR CALL, DATED

70%

RESOLVES

2032-12-31

Supply is byproduct-locked. Over 90 percent of hafnium is tied to nuclear-grade zirconium sponge at roughly 1:50, so output cannot respond to hafnium price on its own, separation capacity sits in a few qualified plants with multi-year qualification, and recycling is about 1 percent. Demand is inelastic on all three legs with no substitute. HfO₂ (and the emerging HfZrO superlattice) has no cheap non-hafnium drop-in at advanced nodes, hafnium's creep contribution in single-crystal blades is metallurgically specific, and its neutron cross-section makes it a deliberate control-rod choice. Three inelastic demand curves on one inelastic byproduct supply is the textbook squeeze, and the substitution frontier raises hafnium intensity rather than relieving it.

WHY IT IS NOT PRICED YET

The bare China-controls-hafnium fact is now priced, with spot tripled to roughly \$13,115/kg by April 2026. What is not priced is the long-horizon structural lock. Hafnium supply is hard-pinned to a nuclear-grade zirconium byproduct ratio that cannot expand to meet demand, while three separate inelastic buyers (reactors, advanced chips, gas turbines) collide on the same trickle and the substitution frontier raises hafnium intensity. There is no liquid hafnium equity or traded multi-year curve discounting this persistence, so the unaccounted-for call is the duration and the cross-sector convergence on a byproduct-locked sub-node, not the single export-control headline.

WHERE THE PRICE SITS TODAY

The live anchor confirms the channel is already moving: ~\$4,365/kg (Jan 2025) to ~\$9,500/kg (Jan 2026) to ~\$13,115/kg (April 2026), driven by China's H2-2025 curbs (exports down ~90 percent) plus AI-compute chip

demand. The directional move is partly priced into spot, which is the main reason the dated call sits below our conviction in the thesis. The unpriced residual is multi-year persistence: thin, illiquid metal with no equity proxy and no forward curve discounting a structural multi-sector squeeze through 2032.

THE BINDING CONSTRAINT

Nuclear- and electronic-grade high-purity hafnium: the poison fraction stripped off nuclear-grade zirconium sponge, specifically the contention between control-rod-grade Hf, the semiconductor HfO₂/HfZrO precursor, and superalloy-additive Hf for the same few-hundred-tonne annual byproduct stream. Not "critical minerals," not "hafnium" broadly, but the byproduct-pinned high-purity sub-node that three sectors share.

WHAT WE ARE WATCHING

High-purity hafnium price (\$/kg) and lead time, the spread between nuclear/electronic grade and industrial grade, and the level of Chinese hafnium export licensing/volumes. Anchor (verified June 2026): Jan 2025 ~\$4,365/kg, Jan 2026 ~\$9,500/kg (+117 percent), ~\$13,115/kg by April 2026; China unwrought hafnium exports down ~90 percent across 2025 (~5,001 kg in Jan to ~499 kg in Sep). Nuclear/electronic grade trades a large premium over industrial.

WHAT WOULD PROVE US WRONG

Kill if, before 2033, high-purity hafnium real price falls back below ~\$5,000/kg and holds for a sustained stretch (signaling new dehafniation or recycling capacity actually caught up, or Chinese licensing meaningfully relaxed), OR if a non-hafnium high-k dielectric stack is qualified at a leading logic node AND a hafnium-free single-crystal superalloy reaches turbine-blade qualification, removing two of the three demand legs. Note the current substitution research moves the wrong way for the bear case (Hf-Zr superlattices use more Hf). Persistent multi-year tightness with China-concentrated licensing leverage and price holding well above the 2025 baseline resolves it true.

HOW WE TRIED TO BREAK IT

We tried three ways to kill it. (1) Already priced? The bare China-export-control fact is priced. Hafnium ran from \$4,365/kg in Jan 2025 to \$9,500 by Jan 2026 and on to ~\$13,115/kg by April 2026, so the export shock itself is in the tape. But the real claim here is narrower than that headline. Hafnium has no liquid equity, trades thin and over-the-counter, and carries almost no sell-side coverage. What is NOT in any priced instrument is the multi-year persistence of a byproduct-pinned node being pulled by three independent inelastic buyers at once. Trade desks (MMTA, Argus) describe the squeeze, but there is no traded multi-year curve that discounts it, so the duration is unpriced even though the spot direction is no longer secret. (2) Elastic supply? No. Over 90 percent of hafnium is a byproduct of nuclear-grade zirconium refining at a fixed roughly 1:50 ratio, you cannot make more Hf without overmaking Zr cladding the market does not want, separation sits in a handful of qualified plants, recycling is about 1 percent, and the only new supply (Tanbreez/Greenland concentrate, a Romanian refinery) is 2030-plus and concentrate-stage, not near-term relief. Inelastic confirmed. (3) Demand legs removable? The substitution risk runs the wrong way for the bears. The frontier high-k research is hafnium-zirconium-oxide superlattices, which use MORE hafnium per wafer, not a hafnium-free stack. Superalloy and control-rod roles are physics-specific with no drop-in. It survives.

Why we are making the call PROMOTE. The mechanism passes all three gate tests. Supply is genuinely inelastic: a fixed roughly 1:50 byproduct ratio off nuclear-grade zirconium, a handful of qualified plants,

about 1 percent recycling, and the only new sources (Tanbreez concentrate, a Romanian refinery) sit at 2030 and beyond, so the constraint cannot expand on a hafnium price signal. Demand is locked on three physics-specific legs with no drop-in substitute, and the leading substitution research raises rather than removes hafnium intensity. On the pricing test, this is the one place we docked it. The export-control headline and the spot direction are no longer secret, the price has already tripled to roughly \$13,115/kg, so the structural conviction is high but the easy directional edge is partly priced. What stays unpriced is duration and the cross-sector convergence on one byproduct node, since hafnium has no liquid equity, trades thin, and carries near-zero sell-side multi-year discounting. That is why we put 70 percent on the exact dated version, below our 82 percent conviction in the thesis itself. The structural case is strong, but the dated call is really a bet on multi-year persistence and on price holding well above the 2025 baseline, which carries a real timing-and-relaxation cost (a Chinese license loosening or a chip-cycle demand dip could pull spot back toward the kill line). The input is named and specific, the supply is inelastic, and it survives the challenge, so it graduates, with the honest boundary that the call is now about persistence rather than about discovering an unknown shortage.

P2 · Through 2034 the binding constraint on whole-mammalian-brain connectomes is not the electron microscope or the segmentation model. It is the human-proofread, error-audited gold reference set. Generative dense-segmentation auto-traces neurons faster every year but cannot self-certify, so verified ground-truth wiring becomes the scarce, slow-to-grow asset, with proofreading commanding the majority of every whole-brain project budget.

The boom: Auto-segmentation models trace neurons, but every merge or split is a silent topology error until a human checks it against ground truth, and the model cannot verify itself because the model is what is being checked. As imaging throughput and model quality rise, the gate moves to the one input that does not scale: a human-proofread reference connectome. A mouse brain holds roughly 70-100M neurons and on the order of 100B-1T synapses, each a potential silent error resolved at a roughly fixed human rate. The Wellcome roadmap and independent estimates put whole-mouse-brain proofreading at 0.3-1M person-years and over 90-97% of project cost (about \$10B over 17 years). MICrONS and FlyWire are the existence proofs, and the larval zebrafish reconstruction is still being proofread years after imaging. The trainable expert pool grows linearly at best, and synthetic reference generation is circular: you would train the verifier on the thing it must verify. · Domain: neuroscience / AI-for-science

STRUCTURAL CASE

82%

OUR CALL, DATED

46%

RESOLVES

2034-12-31

Imaging hardware and segmentation models ride exponential cost curves and improve yearly. The proofread reference rides a human-labor curve: dense-neurite topology agreement cannot be synthetically generated without circularity, and the pool trainable to inter-annotator agreement grows linearly. Each larger brain multiplies synapse count super-linearly against the reference budget. This is a physically and demographically locked bottleneck, not a software gap the next model release closes. The 2025 partial-automation results (90% of guided-proofreading value at 80% lower workflow cost) compress the labor but still depend on human gold data, so they move the cost without dissolving the inelastic input.

WHY IT IS NOT PRICED YET

Public and investor narration treats connectomics as a hardware story (faster multibeam EM) and an AI story (better segmentation), framing proofreading as a transient cost AI will erase. The durable inversion is that better models increase the reference set's value because they still need certifying against it. This is unpriced in the relevant sense: there is no spot price, ticker, or sell-side model for verified connectome reference data. The caveat is that inside the field the proofreading-as-gate framing is no longer secret (the Wellcome roadmap, E11 Bio, and the "if it eliminates proofreading" headline all name it openly), so the edge is the durability and threshold, not the existence, of the bottleneck.

WHERE THE PRICE SITS TODAY

No traded spot price or equity coverage exists for proofread connectome reference data. This is a research-budget-allocation reality, not a market-priced asset. The closest live anchors are the Wellcome roughly \$10B/17-year roadmap estimate (>90-97% proofreading) and Rodriques's conditional \$7-10M optical figure that holds ONLY if proofreading is eliminated. Within the field the bottleneck is already openly named, so it is narratively visible to insiders even though unpriced by markets.

THE BINDING CONSTRAINT

Expert-proofread, error-audited gold-standard connectome reference data (synapse-level verified wiring) plus the human annotator pool trained to inter-expert agreement that produces it. Not the multibeam EM scope, not the auto-segmentation model, not GPU compute. The new automation (Autoproof, ConnectomeBench LLMs) still consumes this exact reference set to train and check against, which raises its value rather than removing it.

WHAT WE ARE WATCHING

Share of total whole-brain connectome project budget and calendar time consumed by proofreading/verification (today reported >90-97% on whole-brain-scale efforts), plus cost per verified cubic millimeter and count of trained expert proofreaders. Resolves TRUE if through 2034 at least one whole-mammalian-brain (mouse-scale) program publicly reports proofreading/verification as the explicitly named gating item AND >80% of cost, and verified-reference cost per mm³ has not fallen by more than ~3x.

WHAT WOULD PROVE US WRONG

A self-certifying reconstruction reaches expert-agreement-grade accuracy on dense neurite topology with NO human proofreading (error rate on held-out human-verified volumes statistically indistinguishable from inter-expert agreement). The live optical-barcoding "\$10M mouse brain if it eliminates proofreading" path is the sharpest such threat; OR proofreading drops below 50% of whole-brain budget before 2034 via automation (Autoproof-class methods already cut workflow cost ~80%); OR synthetic/self-supervised reference generation is shown to remove the circularity. Any one dissolves the constraint.

HOW WE TRIED TO BREAK IT

Two live kill-vectors keep the dated trigger honest. (1) Optical barcoding could uniquely label every neuron and eliminate reconstruction-and-proofreading entirely. The author calls it highly speculative and in need of significant further development, but on a 2034 horizon it is a real tail. (2) Partial automation is already eroding the workflow. Autoproof (Sept 2025) delivers 90% of guided-proofreading value at 80% lower cost and attached 200k fragments (4 proofreader-years), and ConnectomeBench LLMs hit 75-85% on split errors. Neither self-certifies and both still depend on the human gold set, so they do not dissolve the constraint, but an 80% workflow-cost cut, if it generalizes, could plausibly pull proofreading's share below the >80% threshold by 2034 even while staying above the 50% hard kill. The thesis survives. The exact threshold-and-date trigger is taxed.

Why we are making the call The structural case survives the challenge. Proofreading is independently confirmed at over 90-97% of whole-mouse-brain budget, the mechanism (silent topology errors a model cannot self-certify) is physical not software, and the named input, expert-proofread gold-reference data plus the linearly-growing trained annotator pool, is made MORE valuable, not less, by better models that must check against it. It is inelastic and not market-priced, so it clears the pricing and supply-elasticity gates. The dated call sits near a coin flip, below our high conviction in the thesis, because the exact >80%-through-2034 threshold is actively under attack: Autoproof-class automation already cuts workflow cost ~80% and could drag the share under 80% (short of the 50% hard kill), and the optical-barcoding elimination path, though speculative, is a live 2034 tail. PROMOTE on the thesis, and mark the boundary as the threshold and timing, not the bottleneck's existence.

P3 · By 2035 sulfuric acid, set by involuntary fossil-fuel desulfurization that shrinks as the world decarbonizes, becomes the explicitly cited binding gate on at least one HPAL battery-nickel or phosphate-fertilizer expansion, with a sustained real price step-change that outlasts the 2026 acute shocks rather than reverting with them.

The boom: Two decarbonizing systems collide on one molecule that nobody makes on purpose. HPAL nickel-cobalt is roughly 75 percent sulfur-dependent by reagent mass and phosphate fertilizer consumes about 60 percent of global sulfuric acid, yet 80-85 percent of sulfur is an involuntary byproduct of refining sour crude and gas, produced because emissions rules force desulfurization, not because anyone wants it. Burn less fossil fuel and that stream peaks and falls while acid demand climbs toward roughly 400 Mt by 2040 (UCL projects a 100-320 Mt shortfall). Supply is set by oil-and-gas throughput, not by sulfur's own price, so the market cannot clear by incentivizing more. The 2026 spike (sulfur up about 70 percent, Huayou cutting HPAL output about 50 percent, Mosaic cutting about 2 Mt of phosphate) is the acute preview. The structural version is the same squeeze with the geopolitics removed and the byproduct base permanently lower. · Domain: critical materials / agriculture / battery supply chain

STRUCTURAL CASE

78%

OUR CALL, DATED

46%

RESOLVES

2035-12-31

About 80-85 percent of global sulfur is recovered involuntarily from hydrocarbon processing. Its volume is a near-fixed function of how much sour oil and gas is refined, with essentially zero price elasticity of supply. Decarbonization shrinks that base. The two largest sinks, phosphate fertilizer (about 60 percent of acid) and battery-metal HPAL (about 75 percent sulfur-intensive), are both structurally growing and substitution-poor. Standing up Frasch or new smelter-acid capacity is slow and capital-heavy, so a widening gap between involuntary supply and transition-driven demand is locked in by chemistry, not sentiment.

WHY IT IS NOT PRICED YET

In 2022 this was an obscure cross-domain seam. By mid-2026 it is no longer obscure: sulfur is up about 70 percent, HPAL and phosphate producers are cutting output on acid cost, and trade desks (Kpler, CleanTechnica, Atlantic Council) openly call it a feedstock crisis cascading through nickel, copper and fertiliser. What remains genuinely unpriced is the distinction between the acute 2026 shock the market expects to normalise (Hormuz, China and Russia export halts) and the sustained structural premium driven by a permanently shrinking byproduct base as fossil fuels decline. The market prices the shock. The locked-in physics of peak byproduct sulfur coupling food and EV supply over a decade is the unpriced seam.

WHERE THE PRICE SITS TODAY

Live anchor June 2026: NE Asia sulfuric acid about 171 USD/MT and Europe about 176 USD/MT in March 2026, with sulfur up roughly 70 percent year-on-year, but acid prices already correcting -4.7 percent Dec-Mar as logistics bottlenecks eased. So spot has spiked on shock and is partially reverting, which confirms the market reads 2026 as acute, not structural. The structural premium the trigger requires is not yet in forward prices.

THE BINDING CONSTRAINT

Non-fossil-byproduct or smelter-sourced merchant sulfuric acid available to HPAL and phosphate plants, and the involuntary byproduct sulfur stream behind it. The inelastic input is the sulfur/acid molecule, not

nickel ore, phosphate rock, leach autoclaves, or fertilizer plant capacity.

WHAT WE ARE WATCHING

Track (1) benchmark sulfur (Vancouver and Middle East) and merchant sulfuric-acid prices in real terms versus the 2024-2025 average, separating the 2026 shock spike from the post-shock baseline; (2) any HPAL battery-nickel or phosphate-fertilizer expansion publicly citing acid or sulfur availability, not ore, capital, or offtake, as the binding constraint; (3) global byproduct sulfur output trend as refining and sour-gas throughput evolve; (4) pace of any new Frasch, pyrite, or smelter-acid capacity. Resolves TRUE if by 2035-12-31 merchant sulfuric acid sustains a greater-than-40-percent real premium over the 2024-2025 average for at least four consecutive quarters in a window that is not attributable to an acute supply shock, AND at least one HPAL or phosphate expansion publicly cites acid or sulfur supply as the binding gate.

WHAT WOULD PROVE US WRONG

Byproduct sulfur stays ample through 2035 because refining and sour-gas throughput do not fall enough to tighten the structural base, and real acid prices revert to within 20 percent of the 2024-2025 average once the 2026 Hormuz, China and Russia shocks clear, OR battery chemistry shifts decisively away from HPAL nickel (LFP and sodium-ion dominance) while phosphate demand plateaus, collapsing the combined acid pull, OR low-cost non-byproduct acid (new Frasch, pyrite, or smelter capacity) scales fast enough to fill the gap.

HOW WE TRIED TO BREAK IT

The hardest refute is that this is no longer mispriced. As of June 2026 the seam is being actively traded and written up. Sulfur is up roughly 70 percent year-on-year, Huayou has cut Indonesian HPAL output about 50 percent explicitly citing acid cost (acid is 65-70 percent of MHP cash cost), Mosaic has cut about 2 Mt of US phosphate on acid cost, and Kpler, CleanTechnica and the Atlantic Council all frame it openly as a feedstock crisis cascading through nickel, copper and fertiliser. So the cross-domain coupling the candidate calls unpriced is, in mid-2026, very much priced. The thesis only survives on one distinction: the market is pricing an acute geopolitical shock (Hormuz closure 28 Feb, China export ban 10 Apr, Russia ban), not the structural decade-scale peak-sulfur decline. Trade desks expect the shock to normalise, and acid prices were already correcting (-4.7 percent Dec-Mar 2026). The unpriced object is therefore the SUSTAINED structural premium after the shock clears, not the existence of tightness. Supply elasticity survives clean: 80-85 percent of sulfur is involuntary hydrocarbon byproduct with near-zero price elasticity, and Frasch or smelter acid is slow and capital-heavy, so the gap cannot clear by incentive. The kill switches are live, not hypothetical: LFP and sodium-ion are eroding the HPAL-nickel pull, and that demand-collapse path is the main thing that could falsify the demand leg before 2035.

Why we are making the call Survives on three legs. Supply is genuinely inelastic: 80-85 percent of sulfur is involuntary hydrocarbon byproduct set by refining throughput, with near-zero price elasticity, and non-byproduct alternatives are slow and capital-heavy. The mechanism is physics and chemistry, not sentiment, since the UCL study quantifies a 100-320 Mt/yr acid shortfall by 2040 against demand near 400 Mt, with HPAL about 75 percent sulfur-intensive and phosphate about 60 percent of acid use, both growing and substitution-poor. And a real seam remains even though the market is now loudly aware of sulfur tightness: in mid-2026 prices reflect an acute geopolitical shock that desks expect to normalise

(acid already correcting -4.7 percent Dec-Mar), not the sustained structural premium driven by a permanently shrinking byproduct base. Our conviction in the thesis is high because the structural case is strong. The dated call sits near a coin flip because it carries a real timing and measurement cost. The citation leg is arguably already near-met (Huayou and Mosaic cited acid cost in 2026), but the sustained-greater-than-40-percent-real-premium-for-four-quarters leg has to clear AFTER the shocks normalise and against live kill switches, chiefly LFP and sodium-ion eroding the HPAL demand pull before 2035.

P4 · US transmission and HVDC buildout in the 2030s binds on the narrow EHV energizing-and-commissioning craft, not the broadly-priced generic electrician shortage. The binding input is the journeyman and commissioning lead who can live-line and energize EHV lines and converter stations, and roughly half the utility line/substation cohort retires by 2030.

The boom: The loud, life-or-death labor story is the generic electrician for data-center fit-out, now broadly covered and largely priced into the journeyman wage series. The unpriced input sits one layer up and narrower: the extra-high-voltage transmission craft. Energizing a 345-765 kV line, commissioning an HVDC converter station, or doing live-line work on the bulk grid is a separate, much smaller trade learned almost entirely on the job over many years, with thin apprenticeship pipelines. Live search confirms a roughly 3:1 demand-to-supply gap for HVDC and power-electronics commissioning skill and a "small and shrinking" pool with hands-on EHV-scale experience. You cannot cross-train a data-center journeyman into an EHV energization lead in one apprenticeship cycle. As interconnection queues clear and HVDC links get funded, the schedule binds on this craft, and EHV commissioning day-rates detach from the general electrician market. The live offset to watch: factory acceptance testing and digital pre-commissioning are actively pulling work off-site and partly relieving the engineer-side bottleneck. · Domain: energy / human capital

STRUCTURAL CASE

72%

OUR CALL, DATED

46%

RESOLVES

2033-12-31

EHV commissioning and live-line competence is tacit, on-the-job knowledge with a 4-7 year journeyman-plus seasoning path and no fast substitute. The cohort is demographically locked: the people retiring through the 2030s were trained in the 1980s-90s grid-build era, and the replacement pipeline was thinned by two decades of low transmission investment. Live search confirms a roughly 3:1 demand-to-supply gap and a small, shrinking pool with hands-on EHV-scale commissioning experience. Headcount cannot be conjured faster than the training cycle.

WHY IT IS NOT PRICED YET

The market prices the generic electrician shortage but treats labor as one undifferentiated pool. The EHV/HVDC commissioning subtrade is invisible in financial channels: no separate wage series, no listed pure-play, buried inside utility O&M and EPC line items. It is surfacing only in recruiter and EPC trade commentary, not in spot prices or sell-side models. FUTURE_MAP has no human-capital or transmission-craft call, and its closest labor call is LFP process engineers, a different domain.

WHERE THE PRICE SITS TODAY

No isolated price channel exists at the input. Generic journeyman wages are BLS-tracked and rising (~\$59.50/hr) and broadly covered, so the generic layer is priced. EHV/HVDC commissioning labor has no distinct wage series, no pure-play equity, and no sell-side model, and the cost lives inside utility O&M and EPC bids. The gap is real but only narratively visible in recruiter/trade sources, which is why the input is mispriced rather than priced.

THE BINDING CONSTRAINT

Extra-high-voltage and HVDC transmission commissioning and live-line craft labor: the journeymen, test and protection engineers, and energization leads who can commission and energize EHV lines, substations, and

HVDC converter stations. Distinct from the general construction-electrician pool, and narrower still than the utility lineman cohort.

WHAT WE ARE WATCHING

Contractor day-rate and lead time for EHV/HVDC commissioning crews versus general journeyman electrician wages; count of transmission/HVDC projects naming energization/commissioning labor (not permitting or equipment) as the schedule-limiting item. Now: general journeyman ~\$59.50/hr and rising, broadly covered; EHV-commissioning labor not separately tracked or priced; ~half the utility line/substation workforce retirement-eligible by 2030; recruiter/EPC sources report ~3:1 demand-to-supply for HVDC commissioning skill.

WHAT WOULD PROVE US WRONG

If by 2033 EHV/HVDC commissioning labor never appears as a named project-schedule limiter (utilities and EPCs keep citing transformer/equipment lead times, GOES, or interconnection rather than commissioning crews), or if factory acceptance testing, digital-substation pre-commissioning, and remote energization collapse the on-site EHV-craft requirement, the call is dead.

HOW WE TRIED TO BREAK IT

Three live attacks. (1) Already priced? No at the input. The generic shortage is in BLS wage series and Fortune-grade coverage, but EHV/HVDC commissioning has no separate wage series, no pure-play equity, and sits inside utility O&M and EPC line items; recruiter and EPC sources (EuroEngineerJobs, Burns McDonnell, Newport) describe a 3:1 demand-to-supply gap qualitatively, not a priced channel. Survives. (2) Elastic? No. 4-7 year tacit seasoning, cohort trained in the 1980s-90s build era, ~half the utility workforce retirement-eligible by 2030 (US DoL), replacement pipeline thinned by two decades of low transmission capex. Headcount cannot outrun the training cycle. Survives. (3) Substituted away? Partial, and this is the real wound. Factory acceptance testing and digital-substation pre-commissioning genuinely shift labor off-site, and CIGRE VSC-HVDC commissioning work explicitly addresses limitations of on-site testing. This is funded and active, exactly the kill line, and it erodes the engineer-side (protection/control/commissioning) bottleneck even though it cannot collapse live-line energization. The input also blurs two distinct cohorts: live-line journeymen and HVDC commissioning/protection engineers. Net: the structural case is strong, and the dated named-limiter trigger is taxed by the automation offset and by equipment lead times still dominating cited limiters.

Why we are making the call Survives the gate: mispriced at the input (priced generic layer, unpriced EHV/HVDC commissioning subtrade), genuinely inelastic (4-7 year tacit path, demographically locked retirement wave, roughly 3:1 demand-to-supply confirmed live), and survives the challenge on substitution because automation cannot collapse live-line energization even if it erodes the engineer side. The dated call sits well below our conviction in the thesis because the dated resolution depends on commissioning labor becoming a *named* schedule limiter by 2033, against active FAT/digital pre-commissioning offsets and equipment lead times that still dominate cited limiters. Strong structural call, honestly a coin flip on the exact measurement test.

P5 · By 2034 the binding constraint on commercial DT fusion migrates one layer upstream of tritium onto blanket-grade enriched lithium-6, whose only proven industrial process (mercury COLEX) is treaty-banned and whose only current producers are Russia and China. The unpriced chokepoint is qualified, Minamata-compliant Li-6 enrichment capacity outside that duopoly, not plasma, magnets, or tritium.

The boom: When private fusion proves net energy and the plasma bottleneck is funded away, rent jumps upstream into the fuel cycle. Every DT plant breeds its own tritium by hitting Li-6 in the blanket, so each needs a large startup inventory of 6Li-enriched lithium (tens to ~100 t of contained Li-6 per plant, versus natural lithium at only 7.5% Li-6). The detail almost nobody prices: the sole industrially proven route, COLEX, runs on thousands of tonnes of mercury and is banned for new use under the Minamata Convention. The West has essentially zero commercial Li-6 supply, only Russia and China produce it, and China is reportedly just buying from Russia. Whoever first fields a Minamata-compliant, non-mercury Li-6 route at industrial scale captures a chokehold on the entire DT fusion industry. · Domain: energy / fusion fuel cycle

STRUCTURAL CASE

78%

OUR CALL, DATED

46%

RESOLVES

2034-12-31

Natural lithium is 7.5% Li-6 / 92.5% Li-7; achieving a tritium breeding ratio above ~1.05 in most blankets needs Li-6 enriched far above natural abundance (30-90%). Isotope separation one mass unit apart is intrinsically slow and capital-heavy. COLEX, the only industrially proven route, needs thousands of tonnes of mercury per ~100 t Li-6 run over roughly a decade (Y-12 scale) and is banned for new use under Minamata. No drop-in replacement is at industrial scale, and non-mercury alternatives sit at lab/pilot. Supply elasticity is therefore near zero over 5-10 years regardless of price, and the dependency sits in series ahead of tritium itself, because the tritium is bred from the Li-6.

WHY IT IS NOT PRICED YET

Mainstream fusion discourse is confinement, high-field magnets, and at most tritium scarcity. The Li-6 layer beneath tritium, that its only industrial process is mercury-based and Minamata-banned and that supply is a Russia/China duopoly, appears in a handful of 2025-26 papers and nascent policy chatter but is absent from equity coverage, sell-side models, and most fusion-investment framing. A textbook one-layer-upstream migration that is not yet priced as a supply-chain chokepoint.

WHERE THE PRICE SITS TODAY

No commercial Western Li-6 spot market exists, so there is no spot price to inspect, and that absence is itself the signal of un-priced scarcity. Live anchors: Russia/China duopoly confirmed (Novosibirsk NCCP is the Li-7 hub, China reportedly buys from Russia); Western industrial-scale Li-6 capacity is effectively zero. SILEX laser enrichment, the obvious tech analog, is at TRL-6 but for URANIUM, not lithium, and is not a Li-6 supplier. ICOMAX returns no commercial-scale evidence. Policy de-risking (DPA Title III, strategic stockpile, DOE-TVA Watts Bar precedent) is only just being proposed in 2026, not funded or priced.

THE BINDING CONSTRAINT

Qualified blanket-grade enriched lithium-6 (Li-6 enriched well above the 7.5% natural abundance, typically 30-90%) produced by a Minamata-compliant, non-mercury industrial process outside Russia and China.

The inelastic input is certified Li-6 enrichment capacity, not lithium metal, not tritium, not the blanket design.

WHAT WE ARE WATCHING

Track: (1) announced allied (non-Russia/non-China) enriched-Li-6 production capacity in t/yr contained Li-6, currently ~0 at industrial scale; (2) DOE or private-fusion fuel-cycle disclosures naming Li-6 supply as an explicit program gate; (3) any enriched-lithium offtake/supply contract signed by a private fusion company at a disclosed premium. TRUE if by 2034-12-31 at least one pilot/demo DT fusion plant publicly cites enriched-Li-6 availability (not plasma, magnets, or tritium) as a binding schedule constraint OR contracts blanket-grade Li-6 at a disclosed premium, AND no non-China/non-Russia plant has reached qualified industrial-scale (>1 t/yr contained Li-6) Minamata-compliant output.

WHAT WOULD PROVE US WRONG

A non-mercury Li-6 enrichment route (ICOMAX, electrochemical, or laser/AVLIS-class) reaches qualified industrial scale (>1 t/yr contained Li-6) in the US/EU/allied bloc before 2034. OR leading commercial fusion approaches converge on aneutronic, non-DT fuels (notably Helion D-He3) so blanket Li-6 demand never binds. OR DT pilot plants slip past 2034, so the fuel-cycle gate has not yet had to bind by the resolution date.

HOW WE TRIED TO BREAK IT

Three live attempts to break it. (1) Already priced: it is not. There is no commercial Western Li-6 market, hence no spot price to carry it; the thesis lives in one 2025-26 peer-reviewed paper (arXiv 2605.04707, Cell Reports Physical Science) and a few 2026 policy/substack notes proposing DPA Title III action, not in sell-side fusion models. The lead private players (CFS, Helion) are unlisted, so no equity channel prices it either. Genuinely mispriced. (2) Elastic supply: no. Separating two isotopes one mass unit apart is intrinsically slow and capital-heavy; the only proven route is treaty-banned; alternatives are described as taking decades to scale. Elasticity is near zero over 5-10 years. (3) The structure survives, but the exact dated version faces three independent costs that hold our probability on it near a coin flip: the Helion D-He3 branch needs no Li-6 blanket and is fast and well-funded; a DT first-blanket fill could slip past 2034; and a US crash program (DPA Title III is already floated in 2026) could plausibly stand up >1 t/yr allied output before 2034, voiding the AND condition. The structural value-migration call is right, and the exact-by-2034 wording is a timing-and-branch bet on top of it.

Why we are making the call Survives all three refute axes on structure: genuinely mispriced (no spot market, no equity coverage, lives in one 2025-26 paper plus early policy chatter), genuinely inelastic (one-mass-unit isotope separation, the only proven route treaty-banned, alternatives decades out), and a clean one-layer-upstream value migration from plasma to tritium to the boring Li-6 enrichment plant. Live sources corroborate the Russia/China duopoly, the Minamata kill of COLEX, the tens-to-hundreds-of-tonnes demand, and the absence of Western industrial supply. Our conviction in the thesis is high (we put 78 percent on it) because the structural mechanism is physically locked. We hold the exact dated version near a coin flip (46 percent, below that conviction) because the exact-by-2034 wording carries three independent timing and branch costs: the Helion D-He3 non-DT branch never invokes a Li-6 blanket, a DT first-blanket fill can slip past 2034, and a funded US crash program (DPA Title III already floated in 2026) could stand up >1 t/yr allied output before 2034 and void the AND condition. PROMOTE the thesis, and do not inflate the dated trigger.

P6 · By 2033 the binding constraint on the developed-world red-cell supply flips from donation willingness to donor-cohort biology, and O-negative red cells specifically enter chronic structural shortage, with at least one large national blood service (NHSBT, US Red Cross/ABC, or Japanese Red Cross) declaring recurring demographically-driven O-neg emergency or restricted-allocation status that is not attributable to a one-off event.

The boom: The math is a demographic identity, not a forecast. People over 65 consume the majority of red-cell units, and that cohort is the already-born boomer wave aging into peak transfusion demand. The donor-eligible band (17-65) is shrinking in absolute terms across Europe, Japan, Korea, and parts of the US, with roughly 16M eligible US donors in 2025 falling toward 8.7M by 2060 and demand projected to cross supply around 2027. The non-obvious layer is O-negative. Its ~7% share is fixed by genetics, it is drawn from the same shrinking pool, and it is the default for unmatched emergency use. You cannot manufacture O-neg donors, and the manufactured escape hatches (enzyme-converted and cultured red cells) are still pre-clinical or at milliliter scale through this horizon. · Domain: biotech

STRUCTURAL CASE

78%

OUR CALL, DATED

46%

RESOLVES

2033-12-31

Blood type is genetically locked at birth, so the ~7% O-negative share cannot be engineered upward in a population. The 17-65 donor-eligible cohort and the 65+ high-demand cohort for the 2030s are already born and their relative sizes are fixed, so boomers aging into peak transfusion demand is a certainty. Donation frequency has a biological ceiling (every 8 weeks). The two manufactured escapes do not scale on this horizon: enzyme-converted universal red cells are pre-clinical (FDA trial design pending as of early 2026) and cultured red cells remain at milliliter, not therapeutic-unit, scale. The aggregate mismatch is a pure demographic identity. The one genuine elasticity is demand-side substitution (LTOWB and RhD-positive transfusion of RhD-negative men and over-50 women), which can blunt but not abolish the O-neg-specific squeeze.

WHY IT IS NOT PRICED YET

The blood-shortage headline is familiar but read as a donor-recruitment and awareness problem fixable by campaigns. The mispriced claim is that the shortfall is a locked demographic identity (donor cohort aging out while consumer cohort ages in) and that the sharpest, least-substitutable point is O-negative, whose share is genetically fixed. Capital and policy still treat this as behavioral, not structural. There is no traded price channel for blood, so the cliff cannot be "already priced" in the equity/spot sense, and the live operational signal (NHSBT Amber alerts, restricted-allocation thresholds) confirms early stress but is not yet framed by anyone as an irreversible demographic identity.

WHERE THE PRICE SITS TODAY

No traded spot price exists for blood. It is a non-market public-service input, so the thesis cannot be "already in spot prices or sell-side models." The closest live anchors are operational scarcity signals: NHSBT group O Amber alert through 2025 and formal restricted-allocation rules below 60 units, and recurring US Red Cross

emergency-shortage appeals. These show the constraint is biting at the margin but is still managed as an inventory/behavioral problem, not priced as a structural O-neg-specific cliff. The framing is still not yet priced.

THE BINDING CONSTRAINT

O-negative whole-blood/red-cell donors within the genetically-fixed ~7% share of a shrinking 17-65 donor-eligible cohort. The inelastic input is universal-donor red cells for unmatched emergency use, not blood supply in aggregate, and the binding scarcity is the O-neg donor count, not donation willingness.

WHAT WE ARE WATCHING

National O-negative inventory days-of-supply and frequency of O-neg-specific emergency/restricted-allocation appeals (US Red Cross / America's Blood Centers, NHSBT, Japanese Red Cross); ratio of 65+ population to 17-65 donor-eligible population. Live anchors (2025-26): NHSBT held group O on Amber alert into 2025 with formal restricted-allocation rules triggering below 60 units; US eligible donors ~16M in 2025 projected to ~8.7M by 2060 with the supply-demand crossover modeled around 2027; enzyme universal-blood (Avivo/UBC) still pre-clinical with FDA on trial design; cultured red cells (RESTORE) at ~10mL doses.

WHAT WOULD PROVE US WRONG

Dead if by 2033 any of: (a) enzyme-converted, cultured, or synthetic-carrier universal red cells reach routine national-scale supply; (b) demand-side substitution closes the gap, specifically broad LTOWB adoption plus RhD-positive transfusion of RhD-negative males and over-50 women becoming standard enough that no major national service reports a recurring demographically-driven O-neg shortage; or (c) older-donor eligibility expansion plus pathogen reduction restores O-neg days-of-supply to non-crisis levels with no recurring O-neg-specific emergency or restricted-allocation declaration tied to demographics.

HOW WE TRIED TO BREAK IT

Two live levers attack the "inelastic" core and pull our probability on the dated call down. First, the demand side for O-negative is more elastic than the candidate claims. National services already manage O-neg demand directly, not just supply: NHSBT and US protocols transfuse RhD-positive red cells to RhD-negative males and to women over 50 when stocks fall, reserving true O-neg only for under-50 women, children, and anti-D patients. That structurally shrinks O-neg demand without any new technology. Second, the emergency/trauma use that the candidate calls non-substitutable is the exact segment moving fastest off O-neg: low-titer O-positive whole blood (LTOWB) is scaling now across US level-1 trauma centers (TROOP across 15 sites, POWeR-MTP, LA-DROP prehospital), explicitly because only ~7-10% of donors are O-neg. So the single sharpest premise, "you cannot substitute the type in an emergency," is already being falsified operationally in 2025-26. What survives refute: the aggregate identity is real and not a campaign artifact. Eligible donors (17-65) fall from ~16M (2025) to ~8.7M by 2060, demand crosses supply around 2027, and the 65+ cohort consuming the majority of red cells is already born. O-neg's ~7% genetic share is genuinely fixed and the technological escape hatches do not arrive on this horizon: enzyme-converted universal blood (Avivo/UBC) is still pre-clinical with FDA on trial design as of early 2026, and cultured red cells (RESTORE) remain at miniature 10mL doses, nowhere near a therapeutic unit let alone national scale. So the structural pressure is real, and the contestable part is whether it shows up specifically as a NAMED, recurring, demographically-attributed O-neg emergency declaration by 2033, given that the same services have a working demand-side release valve they will pull first.

Why we are making the call Survives the gate but with a real timing-and-measurement tax. The structural case is strong (we put 78 percent on it): a locked demographic identity, a genetically-fixed 7% O-neg share, and no scalable manufactured substitute on the horizon. It clears the pricing test, because capital and policy still treat blood shortage as a donor-recruitment/awareness problem, and no market prices the O-neg-specific cliff (blood is a non-traded public-service input, so there is no spot price or sell-side model to already contain it). But our probability on the exact dated version is only a coin-flip plus (46 percent) because the candidate overstates inelasticity on the demand side. The single non-substitutable premise is being operationally falsified right now: LTOWB is scaling across trauma centers and services already release O-neg demand by giving RhD-positive cells to RhD-negative men and over-50 women. Those valves do not erase the supply identity, but they make it plausible that a major service manages the squeeze through chronic restricted-allocation policy without ever issuing a clean, attributable "demographically-driven O-neg emergency" declaration, which is the measurement the trigger requires. Promote as a forward structural call, and hold the trigger odds honest near 50.

Seeds considered and not promoted

These cleared the supply-side test but did not make the cut, usually because there was no clean way to express the trade or the move was already in the price. We list them because we would rather show our work than bury what we dropped.

SEED	PHYSICAL CASE	WHY NOT PROMOTED
Li-6 enrichment, energy/fusion fuel cycle framing (conviction 0.78 / dated call 0.46, resolves 2034)	DT fusion fuel-cycle chokepoint: blanket-grade enriched Li-6, COLEX mercury route treaty-banned, Russia/China duopoly, one-layer-upstream value migration from plasma to tritium to Li-6 enrichment.	Near-duplicate of the two stronger Li-6 calls already selected (P4 and P7 cover the same mechanism). P7 keeps this exact fuel-cycle migration framing with the cleaner mispriced claim, and P4 adds the export-control trigger. Three Li-6 theses would crowd out diverse mechanisms, so this third framing folds into the kept pair.

Each call is dated, with the line that would prove us wrong fixed the day we make it. We do not move the goalposts: a call is superseded, never quietly edited, and graded in public when its date arrives.